

DSA

- 1) (b) Data and a pointer to the next node
- 2) (c) NULL
- 3) (b) self-referential structure
- 4) (b) $\text{head} == \text{NULL}$
- 5) (b) $\text{newnode} \rightarrow \text{next} = \text{head}; \text{head} = \text{newnode};$
- 6) (b) $\text{temp} = \text{head}; \text{head} = \text{head} \rightarrow \text{next}; \text{free}(\text{temp});$
- 7) (c) Traversing Traversing from ~~head~~ ^{tail} to head
- 8) (b) n
- 9) (c) Copying the data of $p \rightarrow \text{next}$ into p , then deleting $p \rightarrow \text{next}$
- 10) (b) while ($p \rightarrow \text{next} != \text{NULL}$)
- 11) (b) $p \rightarrow \text{next} == \text{NULL}$
- 12) ~~12)~~
- 13) (c) Three
- 14) (b) Exactly one node
- 15) (b) $p = (\text{struct node} *) \text{malloc}(\text{sizeof}(\text{struct node}));$
- 16) (b) Avoid special-case handling for insertion and deletion at the head.
- 17) (c) The first node
- 18) (b) It does not support random (indexed) access
- 19) (b) Loss of access to all nodes, causing a memory leak
- 20) (b) A slow pointer and a fast pointer
- 21) (c) The last node
- 22) (b) Nodes may be scattered anywhere in memory
- 23) $\text{struct node} \{$
 $\text{int data};$
 $\text{struct node} * \text{next};$
};
- 24) 30 50 70
- 25) The line while ($p \rightarrow \text{next}$ correct line
 $\text{struct node} **p = \text{head};$

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